under civilian control, President Eisenhower chose to transform NACA, the long-established federal agency for flight research, into NASA – the National Aeronautics and Space Administration. The second program under the US Air Force was to continue its own reconnaissance satellite program with the CIA, and the third program was controlled by the US Air Force in a military role, while the Army was soon told to end its involvement in space. In 1960 von Braun's team transferred from the Army to NASA, becoming the George C. Marshall Space Flight Center.

NACA had been involved with the USAF in the X-15 experimental rocket plane program, which was regarded by many engineers as a steppingstone on the right road to space travel.

The X-15 could travel almost as fast as a rocket launcher and reach the edge of the atmosphere. NACA engineers had argued that the next generation of X-planes might be able to fly into orbit and fly back down to land. Committed to getting a man into space as quickly as possible, however, NASA had no time to wait and find out whether a winged aircraft could really achieve this. While ready to back von Braun's plan to produce a giant launch rocket for the future, what was needed at this point was a space capsule and a flight plan that would take an American into space and back using existing rocket technology.

By the time NASA was formed, engineer Maxime Faget had already sketched out the requisite vehicle. Small and light enough to be lifted into suborbital flight by a Redstone missile, it would be a cone-shaped capsule equipped with rockets to slow it down before reentry to the atmosphere, and with a heat shield protecting its blunt base. It would then drop into the ocean on parachutes, base first. Despite its limitations, if it meant an American was the first astronaut into space, it would do.

Circus performers were one group initially considered as possible candidates for America's first corps of astronauts, but Eisenhower sensibly laid down that they should be military test pilots.

SPACE APE
NASA sent Ham the chimpanzee into space in January 1961. More fortunate than Soviet space dog Laika, Ham survived the experience.

ASTRONAUT TRAINING

WEIGHTLESS MOVES
Trainee Mercury astronauts undergo the experience of weightlessness in the cargo hold of a transport aircraft. Flown in a vertical parabola, the airplane could create gravity-free conditions for up to 45 seconds as it passed over the top of the arc.

OCEAN EXERCISES
Gemini astronauts come to terms with a "splashdown", the clumsy but fully practical solution to landing without wings.

The training the first American astronauts underwent consisted primarily of exposure to repeated simulations of the experience of space flight, including weightlessness and the g forces produced by rapid acceleration. The experience of weightlessness was created by taking the astronauts up in aircraft that were flown in a parabolic curve, or hump, a trajectory that created shortlived zero-gravity conditions. They were also whirled around at the end of the giant arm of the US Navy's centrifuge in Johnsville, Pennsylvania, which was computer programmed to recreate the different g forces they would experience at various stages of a space flight.

In other exercises the astronauts were made to lie for hours in a simulator at Cape Canaveral, sitting in a seat that had its back on the floor, staring up at a replica of the Mercury spacecraft console. There they were put through the routines of the flight and confronted with a variety of emergencies.

All highly skilled test pilots, the astronauts were initially discontented with a program that seemed to place them in the role of "redundant components" in an automatic system. They wanted to fly their space craft, not just serve as passive guinea pigs to test the effects of space flight on the human body. From the start, the astronauts fought successfully to win more control than the early Soviet cosmonauts enjoyed. They were soon to find themselves playing a vital hands-on role, as automatic systems proved considerably less than one hundred percent reliable.

The Mercury 7

A rigorous selection procedure was set in motion to find the men who would represent the strength and pride of the United States. Candidates were required to have educational qualifications, be in peak physical and psychological condition, and have the sort of lifestyle and personal qualities that would fit them to be American heroes. The seven eventually chosen were all married men with children, between 32 and 37 years old. The project that they joined was dubbed Mercury, and they were the Mercury 7.

The Mercury astronauts did not, of course, possess the uniform qualities that their selectors had perhaps hoped for. Marine Colonel John Glenn, the oldest of the group and a veteran of World War II and Korea, was a man of modest manners and strong religious principle. Alan Shepard and Walter Schirra were wilder characters, with a taste for practical jokes and fast cars. Some of the men's family lives were not quite as near the ideal as their *Life* magazine profiles suggested. But all had the necessary courage and patriotism for the risky task at hand.